

# PROXIES IN NETWORKING

A Smart Middleman Between Users and Servers



Forward Proxy



Reverse Proxy



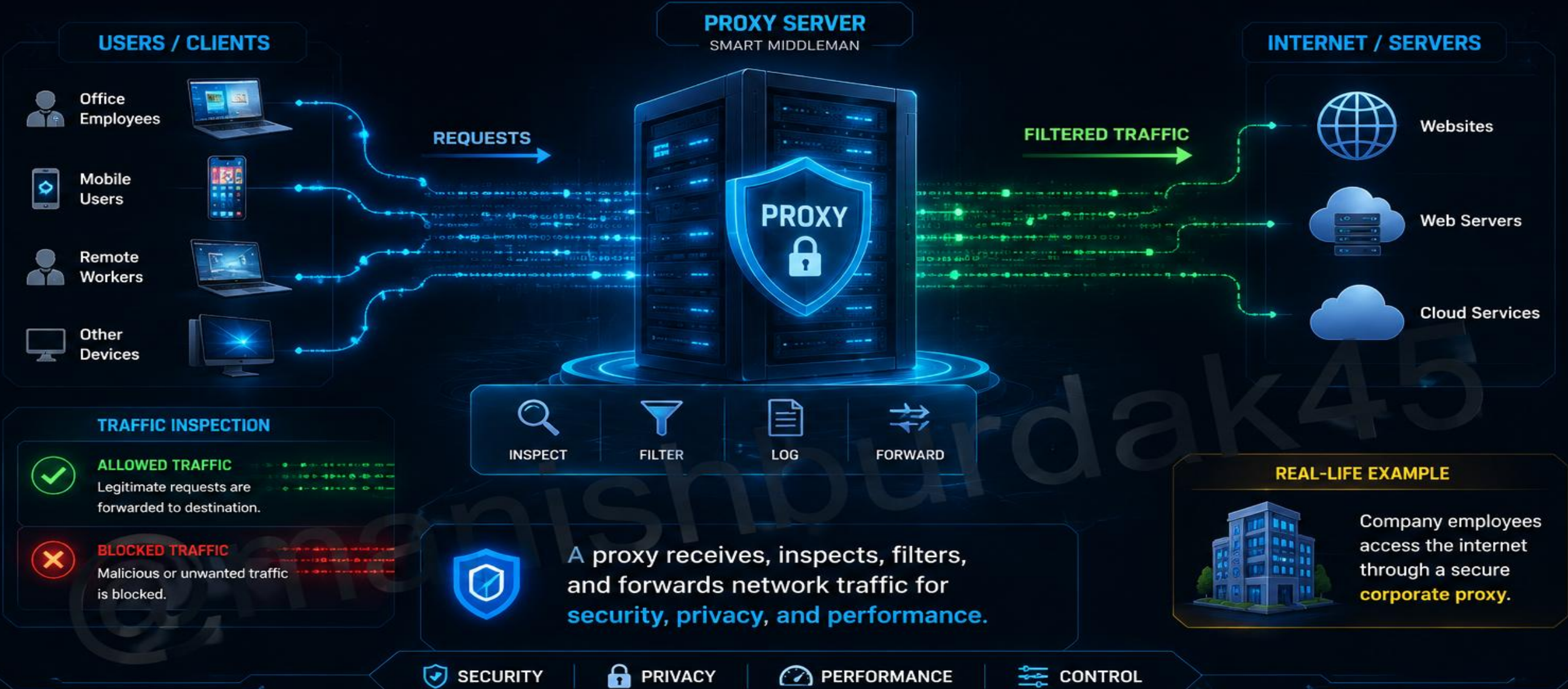
Traffic Filtering



Security Layer



Layer 7 Proxy



# 02 TYPES OF PROXIES

## Two Main Types of Proxies in Networking

### FORWARD PROXY

Acts on behalf of clients.  
Hides client IP and fetches resources from the internet.



### REVERSE PROXY

Acts on behalf of servers.  
Hides server IP and protects backend servers from direct client requests.



### REAL-LIFE EXAMPLE



In a company, employees access the internet through a proxy server. The company can monitor, filter, and control internet usage.

### REAL-LIFE EXAMPLE



CLOUDFLARE

Cloudflare acts as a reverse proxy for websites. It protects the server, hides real IP, and blocks malicious traffic.

### KEY DIFFERENCE



Focuses on Clients



Hides Client IP



Controls & Filters Outgoing Requests



Focuses on Servers



Hides Server IP



Protects & Secures Incoming Requests



Both proxies improve security, privacy, performance, and control network traffic.

# FORWARD PROXY

Works on behalf of clients.

A Forward Proxy sits between users (clients) and the internet. It forwards client requests, **hides the real IP**, filters content, and improves security.

## HOW FORWARD PROXY WORKS



## KEY BENEFITS



Hides Real IP



Content Filtering



Improves Security



Saves Bandwidth



Caching for Better Speed



Access Control & Monitoring

## REAL-LIFE EXAMPLE

In a company or school, all employees access the internet through a forward proxy. The company can block social media, track usage, and protect users.



## COMMON USES



Web Filtering  
Block or allow websites.



Access Control  
Allow only permitted websites.



Bandwidth Saving  
Cache content to save bandwidth.



Anonymity  
Hide client identity and IP address.



Forward Proxy = Client Protection, Control, and Smart Internet Access.

# REVERSE PROXY

Works on behalf of servers.

A Reverse Proxy sits between the internet and backend servers. It receives client requests, **hides the real server IP**, **distributes traffic**, and **protects servers** from attacks.

## HOW REVERSE PROXY WORKS

### USERS / INTERNET



Clients send requests to website

### REQUEST

https://yourwebsite.com

### RESPONSE

Website Content

### REVERSE PROXY SERVER



Reverse Proxy receives requests, applies rules, and forwards to servers.

### FORWARDED REQUEST

(Hides client IP)

### SERVER RESPONSE

(Sent back to Proxy)

### BACKEND SERVERS



Web Server



App Server



Database Server

Servers process request and send response.

## KEY BENEFITS



Hides Server IP



Protects from Attacks



Load Balancing



Improves Performance



Handles SSL / TLS Encryption



Caching for Faster Delivery



Better Monitoring & Control

## REAL-LIFE EXAMPLE

Cloudflare acts as a reverse proxy for websites. It protects the server, hides real IP, and blocks malicious traffic.



CLOUDFLARE

Users / Internet



Your Origin Server



✓ Hides real server IP

✓ Blocks attacks

✓ Improves performance

✓ SSL/TLS encryption

## COMMON USES



Load Balancing

Distribute traffic across multiple servers.



Security & Protection

Block attacks and hide server IP.



SSL/TLS Offloading

Handle SSL encryption at proxy level.



Caching

Store content and deliver faster.



DDoS Protection

Absorb and filter malicious traffic.



Reverse Proxy = Server Protection, Load Balancing, Security, and Better Performance.

# TRANSPARENT vs NON-TRANSPARENT PROXY

Two ways proxies interact with clients.



Transparent Proxy works **without** modifying client settings. Clients are unaware that a proxy is being used. Ideal for filtering, caching, and monitoring.



Non-Transparent Proxy **requires** manual configuration. Clients are aware they are using a proxy. Used for privacy, security, and controlled access.

## TRANSPARENT PROXY



### HOW IT WORKS

- No client configuration needed.
- Proxy intercepts traffic automatically.
- Clients don't know they are using a proxy.
- Common in corporate networks.

## NON-TRANSPARENT PROXY



### HOW IT WORKS

- Client must configure proxy manually.
- Client knows they are using a proxy.
- Offers more control and authentication.
- Used for privacy and restricted access.

## KEY DIFFERENCES

| ASPECT               | TRANSPARENT PROXY              | NON-TRANSPARENT PROXY                  |
|----------------------|--------------------------------|----------------------------------------|
| Client Configuration | Not Required                   | Required                               |
| Client Awareness     | Unaware                        | Aware                                  |
| Visibility           | Hidden                         | Visible                                |
| Use Case             | Filtering, Caching, Monitoring | Privacy, Security, Access Control      |
| Common In            | Corporate, Schools, ISPs       | Individuals, Businesses, Secure Access |

## REAL-LIFE EXAMPLES



### Transparent Proxy Example

A school blocks YouTube using a transparent proxy. Students don't need any settings. The proxy automatically filters content.



### Cloudflare (Both Types)

Cloudflare uses transparent proxying for DDoS protection and CDN. It also offers non-transparent proxy for advanced configurations.



### Corporate Network

Companies use transparent proxies to filter websites, block malware, and monitor employee internet usage.



### VPN / Privacy Users

Individuals configure non-transparent proxies or VPNs manually to hide their IP and access geo-restricted content.



Transparent proxies = Silent protection. Non-transparent proxies = Controlled access.

Choose the right proxy for the right purpose!



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# HOW PROXIES WORK – THE FLOW

A proxy acts as an intermediary between the client and the server.

## STEP-BY-STEP FLOW OF A FORWARD PROXY

### 1 CLIENT REQUEST

The client (user/device) sends a request to access a website.



Users / Clients

### 2 REQUEST TO PROXY

The request is sent to the proxy server first.



PROXY SERVER

### 3 PROXY PROCESSING

The proxy checks, filters, and applies rules (security, caching, access control).

#### PROXY ACTIONS

- Check rules
- Filter content
- Scan for threats
- Cache (if available)
- Hide real IP

### 4 REQUEST TO INTERNET

The proxy forwards the request to the target server on the internet.



INTERNET  
(example.com)

### 5 SERVER RESPONSE

The server sends the response back to the proxy server.



TARGET SERVER  
(example.com)

### 6 RESPONSE TO CLIENT

The proxy forwards the response back to the client.



The client and server never communicate directly.  
The proxy sits in between and manages the entire communication.

## BENEFITS OF USING PROXIES



Improved Security



Privacy & Anonymity



Better Performance



Access Control



Bandwidth Savings



Monitoring & Compliance

## REAL-LIFE EXAMPLE

### At a Company

When an employee tries to open a website like news.com, the request goes to the company proxy first.



### Why?

- To apply security policies
- To block harmful or distracting websites
- To monitor and control internet usage
- To cache and speed up access

## COMMON FUNCTIONS



**Filter Content**  
Block or allow websites and content.



**Access Control**  
Allow only authorized users.



**Threat Protection**  
Scan and block malicious traffic.



**Caching**  
Store content to speed up access.



**Logging & Monitoring**  
Track user activity and bandwidth.



Proxies improve security, privacy, performance, and control by acting as a smart middleman.

@manishburdak45

## 07

# CYBER HYGIENE – EVERYDAY HABITS, BIG IMPACT

Simple daily habits can stop most cyber attacks.



Cyber hygiene is your first line of defense.  
Good habits. Strong security.



Everyone in the organization plays a role.  
Small actions today prevent big problems tomorrow.

## DAILY CYBER HYGIENE HABITS



### STRONG PASSWORDS

Use long, unique passwords with a mix of letters, numbers and symbols.



### ENABLE MFA

Turn on Multi-Factor Authentication for all important accounts.



### BEWARE OF PHISHING

Don't click on suspicious links or open unexpected attachments.



### KEEP SOFTWARE UPDATED

Regular updates fix vulnerabilities and keep you protected.



### SECURE YOUR DEVICES

Lock your screen, use strong PINs, and enable device encryption.



### BACKUP REGULARLY

Keep regular backups of important data in a secure location.



### USE SECURE NETWORKS

Avoid public Wi-Fi for sensitive tasks. Use VPN when necessary.



### THINK BEFORE YOU SHARE

Don't share personal or company information on social media.

## IMPACT OF GOOD CYBER HYGIENE



Reduces Risk of Attacks



Protects Data & Privacy



Improves Productivity



Saves Time & Money



Builds a Strong Security Culture



Protects the Organization

## REAL-LIFE EXAMPLE

One employee clicked on a fake invoice email. Attackers gained access and caused a data breach costing thousands.



### Lesson:

A single click can have a huge impact.  
Good habits can stop it.

## QUICK CYBER HYGIENE TIPS



### Be Alert

Stay alert and trust your instincts.



### Verify

When in doubt, verify before you act.



### Protect

Protect your accounts and data.



### Report

Report suspicious activity immediately.



### Repeat

Make cyber hygiene a daily routine.



Good cyber hygiene is not about technology, it's about behavior.

Small Habits. Strong Defense. Big Difference.



# STAY SECURE, STAY AHEAD

Security is not a product, it's a process. Keep learning. Keep protecting.

“ The best defense is a good offense, backed by strong habits and smart choices.

## KEY TAKEAWAYS



### CYBERSECURITY IS EVERYONE'S JOB

Every click, every action, and every decision impacts the security of the organization.



### UNDERSTAND THE RISKS

Know the threats, attack vectors, and how attackers think.



### USE PROTECTIVE MEASURES

Strong passwords, MFA, updates, backups, and secure habits build strong defense.



### STAY VIGILANT

Be alert, think before you click, and verify before you trust.



### CONTINUOUS IMPROVEMENT

Learn, adapt, and stay updated. Cybersecurity is always evolving.



#### PEOPLE

Awareness & Responsibility



#### PROCESS

Policies, Procedures & Best Practices



#### TECHNOLOGY

Tools, Controls & Automation



#### CONTINUOUS

Monitor, Improve & Evolve



When People, Process, and Technology work together, we create a **strong, resilient, and secure** digital future.

## YOUR ACTION PLAN



### Assess

Understand your current security posture.



### Prioritize

Focus on high-impact risks first.



### Implement

Apply best practices and security controls.



### Monitor

Continuously monitor and detect threats.



### Improve

Learn from incidents and keep improving.

## REMEMBER



Attackers only need one mistake.  
**You need to be right every time.**



It's not about if you will be targeted,  
**it's about when.**



Stay curious. Stay informed.  
**Stay secure.**

## USEFUL RESOURCES



NIST Cybersecurity Framework



CISA  
(cisa.gov)



OWASP  
(owasp.org)



SANS Security Awareness



YouTube: Cyber Security Channels



### TOGETHER, WE CAN BUILD A SAFER DIGITAL WORLD.

Thank you for your time and commitment to cybersecurity.

**Stay alert. Stay safe. Stay ahead.**



# 09 NETWORKING MODELS – OVERVIEW

Frameworks that define how data is transmitted from one device to another.



Networking models are layered architectures that standardize communication between systems. They break down the complex process of data transmission into smaller, manageable layers, where each layer has a specific role and responsibility.

## WHY WE NEED NETWORKING MODELS?

- ✓ Ensure devices from different vendors can communicate
- ✓ Provide a standard framework for troubleshooting
- ✓ Simplify the understanding of complex communication
- ✓ Help in designing, analyzing, and securing networks

## THE OSI MODEL (7 LAYERS)

A conceptual model by ISO that defines 7 layers.

|   |  |                           |                                                                    |
|---|--|---------------------------|--------------------------------------------------------------------|
| 7 |  | <b>Application Layer</b>  | Provides network services to user applications. (HTTP, FTP, DNS)   |
| 6 |  | <b>Presentation Layer</b> | Handles data format, encryption, compression. (SSL, TLS, JPEG)     |
| 5 |  | <b>Session Layer</b>      | Manages sessions, establishes, maintains and terminates sessions.  |
| 4 |  | <b>Transport Layer</b>    | Ensures end-to-end delivery, segmentation, reliability. (TCP, UDP) |
| 3 |  | <b>Network Layer</b>      | Logical addressing and routing (IP, ICMP)                          |
| 2 |  | <b>Data-Link Layer</b>    | Node-to-node delivery, framing, error detection. (MAC, Switch)     |
| 1 |  | <b>Physical Layer</b>     | Transmits raw bits over physical media. (Cables, Signals)          |

VS

## THE TCP/IP MODEL (4 LAYERS)

A practical model that forms the foundation of the Internet.

|   |  |                                                                                                             |                             |
|---|--|-------------------------------------------------------------------------------------------------------------|-----------------------------|
| 4 |  | <b>Application Layer</b><br>Combines OSI Application, Presentation & Session layers. (HTTP, FTP, DNS, SMTP) | Maps to OSI: Layers 5, 6, 7 |
| 3 |  | <b>Transport Layer</b><br>End-to-end communication, reliability, flow control. (TCP, UDP)                   | Maps to OSI: Layer 4        |
| 2 |  | <b>Internet Layer</b><br>Logical addressing and routing of packets across networks. (IP, ICMP)              | Maps to OSI: Layer 3        |
| 1 |  | <b>Link Layer</b><br>Combines OSI Data-Link and Physical layers. (MAC, Ethernet)                            | Maps to OSI: Layers 1, 2    |



## KEY TAKEAWAY

OSI is a **theoretical reference** model with 7 layers, while TCP/IP is the **practical 4-layer model** used in real networks. Both models help us understand how data flows from source to destination.



**Goal:** Understand the structure before analyzing the flow.

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## KEY TAKEAWAY

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# NETWORKING MODELS – OSI vs TCP/IP

Two reference models that explain how data is communicated over a network.



Both models divide the communication process into layers. Each layer has a specific role and interacts with the layers above and below it to ensure reliable data transfer.



## KEY PURPOSE

- ✓ Standardize communication between systems
- ✓ Simplify complex networking functions
- ✓ Help in troubleshooting and analysis
- ✓ Provide a framework for protocol development

## THE OSI MODEL (7 LAYERS)

Developed by ISO as a conceptual reference model.

|   |  |                           |                                                                                     |
|---|--|---------------------------|-------------------------------------------------------------------------------------|
| 7 |  | <b>Application Layer</b>  | Provides network services to user applications. (HTTP, FTP, DNS, SMTP)              |
| 6 |  | <b>Presentation Layer</b> | Handles data format, encryption, compression and translation. (SSL, TLS, JPEG, PNG) |
| 5 |  | <b>Session Layer</b>      | Manages sessions, establishes, maintains and terminates sessions. (NetBIOS)         |
| 4 |  | <b>Transport Layer</b>    | Ensures end-to-end delivery, segmentation, reliability and flow control. (TCP, UDP) |
| 3 |  | <b>Network Layer</b>      | Logical addressing and routing of data packets. (IP, ICMP)                          |
| 2 |  | <b>Data-Link Layer</b>    | Node-to-node delivery, framing, error detection. (MAC, Switch, Bridge)              |
| 1 |  | <b>Physical Layer</b>     | Transmits raw bits over physical media. (Cables, Signals, Network Card)             |

VS

## THE TCP/IP MODEL (4 LAYERS)

The practical model used in the Internet protocol suite.

|   |  |                                                                                                             |                                      |
|---|--|-------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 4 |  | <b>Application Layer</b><br>Combines OSI Application, Presentation & Session layers. (HTTP, FTP, DNS, SMTP) | <b>MAPS TO OSI</b><br>Layers 5, 6, 7 |
| 3 |  | <b>Transport Layer</b><br>End-to-end communication, reliability and flow control. (TCP, UDP)                | Layer 4                              |
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| 1 |  | <b>Link Layer</b><br>Combines OSI Data-Link and Physical layers. (MAC, Ethernet)                            | Layers 1, 2                          |



## KEY TAKEAWAY

OSI is a **theoretical 7-layer** model used for understanding and analysis, while TCP/IP is a **practical 4-layer model** used in real-world networking.



Use OSI to understand concepts in depth.



Use TCP/IP to implement and troubleshoot.



Both models together make networking effective and reliable.

# 12 TCP vs UDP – TRANSPORT LAYER PROTOCOLS

Two main protocols at Layer 4 that handle end-to-end communication between applications.



## WHAT IS THE TRANSPORT LAYER?

It provides process-to-process communication, uses port numbers, ensures data delivery (TCP) or fast delivery (UDP) between hosts.



## MAIN FUNCTIONS

- ✓ Segmentation of data
- ✓ Port addressing
- ✓ Flow control
- ✓ Error control (TCP)
- ✓ Multiplexing / Demultiplexing
- ✓ Connection management (TCP)

## TCP (Transmission Control Protocol)



Reliable



Connection-Oriented



Ordered Delivery



Error Checking & Recovery



Slower but Accurate

VS

## UDP (User Datagram Protocol)



Fast



Connection-less



No Order Guarantee



No Error Recovery



Lightweight & Efficient

## TCP EXAMPLES

- Web Browsing (HTTP/HTTPS)
- Email (SMTP, IMAP)
- File Transfer (FTP)
- Remote Login (SSH)
- Online Banking



Reliable Delivery  
All data received in correct order

| Yes                                        | Comparison        | No                                     |
|--------------------------------------------|-------------------|----------------------------------------|
| Reliable (Acknowledgement, Retransmission) | Reliability       | Best Effort (No Guarantee)             |
| Maintains Order                            | Reliability       | No Order Guarantee                     |
| Flow Control, Congestion Control           | Order Delivery    | No Flow Control, No Congestion Control |
| Larger Header (20–60 bytes)                | Control Mechanism | Small Header (8 bytes)                 |
| Slower                                     | Speed             | Faster                                 |
| Higher                                     | Overhead          | Lower                                  |

## UDP EXAMPLES

- Online Gaming
- Video Streaming
- Voice over IP (VoIP)
- DNS Queries
- Live Broadcast



Fast Delivery  
Some loss acceptable, speed is important

## TCP 3-WAY HANDSHAKE



Client

1. SYN

2. SYN-ACK

3. ACK



Server

After this handshake, connection is established and data transfer begins.  
After communication, connection is terminated.



## PORT NUMBERS

Both TCP and UDP use Port Numbers to identify applications.

Examples:

- HTTP – 80 (TCP)
- HTTPS – 443 (TCP)
- DNS – 53 (UDP/TCP)
- DHCP – 67/68 (UDP)

## UDP CHARACTERISTICS

- ✓ No connection setup
- ✓ No acknowledgements
- ✓ No retransmission
- ✓ No guarantee of delivery
- ✓ Minimal overhead
- ✓ Ideal for real-time applications



SPEED FIRST

Perfect for real-time communication



## KEY TAKEAWAY

Use **TCP** when reliability and accuracy are important.

Use **UDP** when speed and low latency are important.



The Internet uses both **TCP** and **UDP** based on the requirement.

# TCP vs UDP – RELIABILITY & PERFORMANCE COMPARISON

Choosing the right protocol depends on the application requirement.



## OVERVIEW

TCP is connection-oriented and reliable, ensuring data is delivered accurately but with more overhead.

UDP is connectionless and lightweight, faster but does not guarantee delivery.



## KEY DIFFERENCE AT A GLANCE

- ✓ TCP ensures reliability
- ✓ UDP ensures speed
- ✓ TCP has more overhead
- ✓ UDP has low latency



### TCP

Email, File Transfer, Web Browsing, Online Banking



### UDP

Video Streaming, Online Gaming, VoIP, DNS, Live Broadcast

## TCP FLOW



Client

1. SYN
2. SYN-ACK
3. ACK



Server

Connection Established

## TCP (Transmission Control Protocol)



Connection Oriented



Reliable Delivery



Ordered Delivery



Error Checking & Recovery



Slower Speed



Higher Overhead

VS

## UDP (User Datagram Protocol)



Connectionless



No Delivery Guarantee



No Order Guarantee



No Error Recovery



Faster Speed



Lower Overhead

|                |                                                                        |
|----------------|------------------------------------------------------------------------|
| Connection     | Connection is established before data transfer (3-way handshake)       |
| Reliability    | Reliable (Acknowledgement, Retransmission)                             |
| Order          | Maintains order of data packets                                        |
| Error Handling | Detects errors and recovers lost or corrupted data                     |
| Speed          | Slower due to connection and error handling                            |
| Overhead       | Higher overhead (Header size: 20–60 bytes)                             |
| Use Case       | Web browsing, Email, File transfer, Remote login, Online banking, etc. |

|                  |
|------------------|
| Connection Setup |
| Reliability      |
| Ordering         |
| Error Handling   |
| Speed            |
| Overhead         |
| Use Case         |

|                                                                         |
|-------------------------------------------------------------------------|
| No connection is established (data is sent directly)                    |
| Unreliable (No Acknowledgement, No Retransmission)                      |
| Does not maintain order of data packets                                 |
| No error detection or recovery                                          |
| Faster (Low latency)                                                    |
| Lower overhead (Header size: 8 bytes)                                   |
| Video streaming, Online gaming, VoIP, DNS queries, Live broadcast, etc. |

## UDP FLOW



Sender



Data (No Handshake)



Receiver

No Connection Direct Transfer

## WHEN TO USE WHAT?



Use TCP when reliability, accuracy and data integrity are critical.



Use UDP when speed, low latency and real-time performance are critical.



## PERFORMANCE IMPACT

TCP ensures accuracy with higher latency.  
UDP ensures speed with minimal latency.  
Choose based on what your application values most!



## KEY TAKEAWAY

There is no one-size-fits-all. Understanding the difference helps in building efficient and effective network applications.



The Internet uses both TCP and UDP based on the application requirements.

# TCP vs UDP - USE CASES & APPLICATION SCENARIOS

Different applications have different needs. TCP and UDP are designed to serve different use cases.



## TCP - RELIABLE & CONNECTION-ORIENTED

- Ensures complete & accurate data delivery
- Uses more bandwidth and has higher overhead
- Slower due to connection setup and error handling
- Best for applications where accuracy is critical



## UDP - FAST & CONNECTIONLESS

- No connection setup, faster transmission
- Lower overhead, uses less bandwidth
- No guarantee of delivery or order
- Best for applications where speed is critical



## QUICK RULE

If accuracy & reliability matter more → **Use TCP**  
If speed & low latency matter more → **Use UDP**

## USE CASES FOR TCP

|                       |                                                             |                                  |
|-----------------------|-------------------------------------------------------------|----------------------------------|
| Web Browsing          | Reliable delivery of web pages, images, scripts.            | HTTP/HTTPS                       |
| Email                 | Ensures complete and correct delivery of emails.            | SMTP, IMAP, POP3                 |
| File Transfer         | Files must be delivered completely and without errors.      | FTP, SFTP                        |
| Remote Login          | Requires reliable communication for commands and responses. | SSH, Telnet                      |
| Online Banking        | Security and accuracy are critical for transactions.        | HTTPS, TCP                       |
| IoT Device Management | Commands and updates must be received reliably.             | MQTT (over TCP), CoAP (over TCP) |

## USE CASES FOR UDP

|                             |                                                            |                                       |
|-----------------------------|------------------------------------------------------------|---------------------------------------|
| Video Streaming             | Can tolerate some packet loss for smooth playback.         | YouTube Live, Netflix Live, IPTV      |
| Online Gaming               | Low latency is critical for real-time player interactions. | PUBG, Call of Duty, Fortnite          |
| Voice over IP (VoIP)        | Real-time voice, slight loss is acceptable.                | WhatsApp Call, Zoom Call, Skype Call  |
| DNS Lookups                 | Quick queries and responses are important.                 | DNS (UDP Port 53)                     |
| Live Broadcast              | Sending to many users at once, speed is priority.          | Live Sports, News Broadcast           |
| IoT Sensor Data (Real-time) | Small, frequent data updates, loss can be tolerated.       | Temperature, Location, Motion Sensors |

## REAL-WORLD EXAMPLES

- Opening a website → TCP (ensures full page loads)
- Playing an online game → UDP (reduces lag)
- Watching a live match → UDP (smooth streaming)



## KEY INSIGHT

TCP is like a courier with **delivery confirmation**.  
UDP is like a **postcard** – fast but no confirmation.  
Choose the right protocol based on what your application values most: accuracy or speed.



## PERFORMANCE TRADE-OFF

### TCP

More Reliable  
More Overhead  
Higher Latency



### UDP

Less Reliable  
Less Overhead  
Lower Latency



Both **TCP** and **UDP** are essential for the Internet – understanding when and where to use them is key to building efficient applications.

# TCP vs UDP – SUMMARY & KEY TAKEAWAYS

Two powerful protocols with different strengths – choose the right one for your application.



## REMEMBER

There is no one-size-fits-all.  
The right choice depends on your application needs.

| TCP (Transmission Control Protocol)                                               |                                                                      | FEATURE       | UDP (User Datagram Protocol)                                                        |                                                                 | COMMON USE CASES                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                  |                                                                                                  |                                                                                                          |                                                                                               |                                                                                                        |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
|   | Connection-oriented.<br>3-way handshake is used.                     | Connection    |   | Connectionless.<br>No handshake required.                       | <br><b>USE TCP WHEN...</b> <ul style="list-style-type: none"><li>Accuracy and reliability are critical (e.g., emails, file transfers, banking)</li><li>Data must be delivered completely and in correct order</li><li>Connection setup and slight delay are acceptable</li></ul> | <br><b>USE UDP WHEN...</b> <ul style="list-style-type: none"><li>Speed and low latency are critical (e.g., gaming, streaming, live events)</li><li>Some packet loss is acceptable</li><li>Real-time data delivery is more important</li></ul> |                                                                                                  |                                                                                                          |                                                                                               |                                                                                                        |
|   | Reliable delivery.<br>Uses ACKs, retransmissions.                    | Reliability   |   | Unreliable delivery.<br>No ACKs or retransmissions.             |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                  |                                                                                                  |                                                                                                          |                                                                                               |                                                                                                        |
|   | Maintains order of packets.<br>Data is delivered in sequence.        | Ordering      |   | No order guarantee.<br>Packets may arrive out of order.         |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                  |                                                                                                  |                                                                                                          |                                                                                               |                                                                                                        |
|   | Has error checking and recovery.<br>Ensures data integrity.          | Error Control |   | No error checking or recovery.<br>Errors are not handled.       |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                  |                                                                                                  |                                                                                                          |                                                                                               |                                                                                                        |
|   | Has flow control.<br>Prevents sender from overwhelming the receiver. | Flow Control  |   | No flow control.<br>Sender may overwhelm the receiver.          | <b>KEY CHARACTERISTICS AT A GLANCE</b>                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                  |                                                                                                  |                                                                                                          |                                                                                               |                                                                                                        |
|   | Slower due to connection setup, reliability checks and overhead.     | Speed         |   | Faster.<br>Minimal overhead.                                    |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                  |                                                                                                  |                                                                                                          |                                                                                               |                                                                                                        |
|   | Higher overhead.<br>Larger header (20–60 bytes).                     | Overhead      |   | Lower overhead.<br>Small header (8 bytes).                      |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                  |                                                                                                  |                                                                                                          |                                                                                               |                                                                                                        |
|  | Best for applications where accuracy and reliability are critical.   | Use Case      |  | Best for applications where speed and low latency are critical. |                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                  |                                                                                                  |                                                                                                          |                                                                                               |                                                                                                        |
|                                                                                   |                                                                      |               |                                                                                     |                                                                 | <b>TCP</b>                                                                                                                                                                                                                                                                                                                                                          | <br>Reliable                                                                                                                                                                                                                                  | <br>Ordered   | <br>Error Recovery    | <br>Slower | <br>Higher Overhead |
|                                                                                   |                                                                      |               |                                                                                     |                                                                 | <b>UDP</b>                                                                                                                                                                                                                                                                                                                                                          | <br>Unreliable                                                                                                                                                                                                                                | <br>Unordered | <br>No Error Recovery | <br>Faster | <br>Lower Overhead  |

## KEY TAKEAWAY



TCP is like a safe, reliable courier that ensures your package reaches the destination correctly.

UDP is like a fast delivery service that gets your package there quickly, without guarantees.

## CHOICE GUIDE



Choose **TCP** for reliability.  
Choose **UDP** for speed.  
Choose smartly for better performance and user experience.

## FOCUS ON YOUR GOAL



Understand your application requirements first.  
Then select the protocol that aligns with your performance, reliability and user experience goals.

## BOTH ARE ESSENTIAL



The Internet needs both TCP and UDP.  
They work together to power billions of applications and services worldwide.



Understanding **TCP** and **UDP** helps you design better applications, optimize performance and deliver better experiences to your users.

# 16 TCP vs UDP – PROTOCOL SCENARIOS & BEST PRACTICES

Pick the right protocol for the right scenario to build efficient, reliable and high-performance applications.



## QUICK RULE

If your application can tolerate some loss and needs speed → **Use UDP**  
If your application needs reliability and order → **Use TCP**



RELIABLE &  
ACCURATE

## WHEN TO USE TCP

- ✓ When accuracy, reliability and order are critical
- ✓ When all data must be delivered correctly
- ✓ When connection setup and retransmissions are acceptable
- ✓ When slight delay is okay for complete delivery
- ✓ Examples: Web, Email, File Transfer, Online Banking

VS

## WHEN TO USE UDP

- ✓ When speed and low latency are critical
- ✓ When some packet loss is acceptable
- ✓ When real-time data delivery is more important
- ✓ When connectionless communication is sufficient
- ✓ Examples: Streaming, Gaming, VoIP, Live Broadcast



FAST &  
LIGHTWEIGHT

## APPLICATION SCENARIOS COMPARISON

| APPLICATION SCENARIO | DESCRIPTION                          | BETTER PROTOCOL |     | WHY?                                                     |
|----------------------|--------------------------------------|-----------------|-----|----------------------------------------------------------|
|                      |                                      | TCP             | UDP |                                                          |
| Web Browsing         | Loading websites, images, scripts    | ✓               | ✗   | Needs reliable delivery of all web content.              |
| Email                | Sending and receiving emails         | ✓               | ✗   | Emails must be delivered completely and in order.        |
| File Transfer        | Uploading/downloading files          | ✓               | ✗   | File integrity and complete delivery are critical.       |
| Online Gaming        | Real-time multiplayer gaming         | ✗               | ✓   | Low latency is critical, some packet loss is acceptable. |
| Video Streaming      | Live or on-demand video streaming    | ✗               | ✓   | Speed and smooth playback are more important.            |
| Voice over IP (VoIP) | Voice calls over the Internet        | ✗               | ✓   | Real-time voice delivery needs low latency.              |
| Live Broadcast       | Live sports, news, events streaming  | ✗               | ✓   | Timely delivery to many users, some loss is okay.        |
| Online Banking       | Transactions, account operations     | ✓               | ✗   | Security, reliability and accuracy are critical.         |
| DNS Lookup           | Resolving domain names               | ✗               | ✓   | Fast query/response, small packets, connectionless.      |
| IoT / Sensor Data    | Sending small, frequent data updates | ✗               | ✓   | Low overhead and fast transmission are important.        |

## BEST PRACTICES



**Understand your application requirements first:**  
Decide based on reliability, accuracy, speed and latency needs.



**Use TCP for critical data:**  
When data integrity and complete delivery are non-negotiable.



**Use UDP for real-time applications:**  
When performance and low latency are more important than perfect delivery.



**Test and monitor:**  
Always test your application under real network conditions.



**Optimize accordingly:**  
Tune buffer sizes, timeouts and retransmission settings as per protocol and use case.



**Know your users:**  
Choose the protocol that delivers the best experience for your users.



## KEY TAKEAWAY



**TCP** is your choice for reliability, accuracy and ordered delivery.



**UDP** is your choice for speed, low latency and real-time performance.



There is no one-size-fits-all – choose the protocol that fits your application needs.



Understanding the difference helps you build better, faster and more reliable applications.



Both **TCP** and **UDP** power the Internet. The right choice depends on your application, your users and your goals.

@manishburdak45

Different design choices make TCP and UDP suitable for different situations.

### TCP (Transmission Control Protocol)

Connection-oriented, reliable, ordered and congestion-controlled.



#### Connection-Oriented

3-way handshake before data transfer.



#### Reliable Delivery

Uses ACKs and retransmissions.



#### Ordered Delivery

Maintains sequence and delivers in order.



#### Error Control

Detects errors and retransmits data.



#### Flow Control

Prevents sender from overwhelming receiver.



#### Congestion Control

Adjusts sending rate to avoid congestion.

VS

### UDP (User Datagram Protocol)

Connectionless, lightweight, fast and best-effort delivery.



#### Connectionless

No handshake. Send data directly.



#### Best-Effort Delivery

No guarantee of delivery or order.



#### Unordered Delivery

Packets may arrive out of order.



#### No Error Recovery

No retransmission. Errors are ignored.



#### No Flow Control

Sender can send as fast as possible.



#### Lightweight

Small header (only 8 bytes).

### FEATURE COMPARISON

| FEATURE            | TCP                                      | UDP                                | WINNER          |
|--------------------|------------------------------------------|------------------------------------|-----------------|
| Connection         | Connection-oriented                      | Connectionless                     | Depends on need |
| Reliability        | Reliable (ACK + Retransmission)          | Best-effort (No guarantee)         | TCP             |
| Ordering           | Ordered delivery                         | Unordered delivery                 | TCP             |
| Error Handling     | Detects & recovers errors                | No error recovery                  | TCP             |
| Flow Control       | Yes (Receiver based)                     | No                                 | TCP             |
| Congestion Control | Yes                                      | No                                 | TCP             |
| Speed / Latency    | Slower (More overhead)                   | Faster (Low latency)               | UDP             |
| Header Size        | Larger (20–60 bytes)                     | Smaller (8 bytes)                  | UDP             |
| Bandwidth Overhead | Higher                                   | Lower                              | UDP             |
| Use Cases          | Web, Email, File Transfer, Banking, etc. | Streaming, Gaming, VoIP, DNS, etc. | Depends on need |

### HEADER STRUCTURE COMPARISON

#### TCP HEADER (20–60 Bytes)

|                            |              |                       |                     |
|----------------------------|--------------|-----------------------|---------------------|
| Source Port (16)           |              | Destination Port (16) |                     |
| Sequence Number (32)       |              |                       |                     |
| Acknowledgment Number (32) |              |                       |                     |
| Data Offset (4)            | Reserved (3) | Flags (9)             | Window Size (16)    |
| Checksum (16)              |              |                       | Urgent Pointer (16) |
| Options (0–40 bytes)       |              |                       | Padding             |

+ Data (Variable)

#### UDP HEADER (8 Bytes)

|                  |                       |
|------------------|-----------------------|
| Source Port (16) | Destination Port (16) |
| Length (16)      | Checksum (16)         |

+ Data (Variable)

### PERFORMANCE IMPACT



- TCP ensures accuracy and reliability at the cost of speed and overhead.
- UDP delivers speed and low latency with minimal overhead but without guarantees.

### DESIGN CHOICE



- Choose TCP when data integrity and reliability are critical.
- Choose UDP when speed and real-time delivery are more important.

### KEY INSIGHT



TCP is like a secure delivery service – slower but reliable.  
UDP is like a postcard – fast but no confirmation.

### REMEMBER



There is no one-size-fits-all. The right protocol depends on your application requirements, user experience and performance goals.



Understanding the strengths and weaknesses of TCP and UDP helps you design better, faster and more reliable applications.

@manishburdak45

Understanding how TCP and UDP behave in real-world conditions to choose the right protocol.



## TCP – RELIABLE BUT HEAVIER

- Ensures every byte reaches the destination.
- Handles errors, retransmissions and congestion automatically.
- Slower due to handshakes, acknowledgments and overhead.
- Best for accuracy, reliability and ordered delivery.



## UDP – FAST BUT LIGHTER

- No guarantee of delivery, order or error recovery.
- Very low latency and low overhead.
- Better performance in real-time and high-speed scenarios.
- Best when speed is critical and small loss is acceptable.



## QUICK RULE

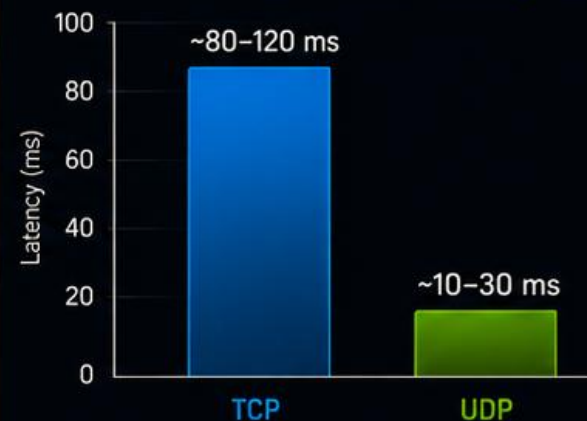
If accuracy & reliability matter more  
→ **Use TCP**

If speed & low latency matter more  
→ **Use UDP**

## PERFORMANCE & RELIABILITY COMPARISON

| ASPECT                | TCP                                     | UDP                                     | WINNER     |
|-----------------------|-----------------------------------------|-----------------------------------------|------------|
| Reliability           | High – ensures delivery                 | Low – no delivery guarantee             | <b>TCP</b> |
| Delivery Order        | Guaranteed in-order delivery            | No order guarantee                      | <b>TCP</b> |
| Error Handling        | Automatic recovery & retransmission     | No recovery – lost data is lost         | <b>TCP</b> |
| Speed / Latency       | Slower – more overhead                  | Faster – minimal overhead               | <b>UDP</b> |
| Overhead              | Higher (headers, acks, handshakes)      | Lower (small header – 8 bytes)          | <b>UDP</b> |
| Bandwidth Efficiency  | Lower – more control data               | Higher – more data, less overhead       | <b>UDP</b> |
| Congestion Control    | Built-in congestion control             | No congestion control                   | <b>TCP</b> |
| Real-time Suitability | Less ideal (variable delay)             | Highly ideal (low & stable delay)       | <b>UDP</b> |
| Security              | More secure for critical transfers      | Less secure (no built-in checks)        | <b>TCP</b> |
| Best Use              | Critical, financial, transactional apps | Streaming, gaming, VoIP, real-time apps | Depends    |

## LATENCY COMPARISON (EXAMPLE)



UDP is faster  
but less reliable.  
TCP is slower  
but more reliable.  
Choose based on  
your application  
requirements.

## BEST PROTOCOL FOR DIFFERENT NEEDS

|                                                       |                                            |                                                          |                                                    |                                              |                                                 |                                             |                                                   |
|-------------------------------------------------------|--------------------------------------------|----------------------------------------------------------|----------------------------------------------------|----------------------------------------------|-------------------------------------------------|---------------------------------------------|---------------------------------------------------|
| Web Browsing<br><b>TCP</b><br>(reliable page loading) | Email<br><b>TCP</b><br>(accurate delivery) | Online Banking<br><b>TCP</b><br>(security & reliability) | Video Streaming<br><b>UDP</b><br>(smooth playback) | Online Gaming<br><b>UDP</b><br>(low latency) | VoIP / Calls<br><b>UDP</b><br>(real-time voice) | Live Broadcast<br><b>UDP</b><br>(low delay) | IoT / Sensors<br><b>UDP</b><br>(small data, fast) |
|-------------------------------------------------------|--------------------------------------------|----------------------------------------------------------|----------------------------------------------------|----------------------------------------------|-------------------------------------------------|---------------------------------------------|---------------------------------------------------|

## KEY TRADE-OFF

| TCP            | UDP           |
|----------------|---------------|
| More Reliable  | Less Reliable |
| More Overhead  | Less Overhead |
| Higher Latency | Lower Latency |



## BOTTOM LINE



Use **TCP** when  
data must be perfect.

Use **UDP** when  
speed is the priority.



Both **TCP** and **UDP** are powerful. The right choice depends on your application, users and performance goals.

@manishburdak45

Different applications have different needs – choose the protocol that fits best.

### WHEN TO USE TCP – REAL WORLD EXAMPLES

| APPLICATION / SCENARIO                                                                                     | WHY TCP IS THE RIGHT CHOICE                                            |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
|  Web Browsing (HTTP/HTTPS) | Web pages, images, and scripts must load completely and accurately.    |
|  Email (SMTP, IMAP, POP3)  | Emails must be delivered completely and in order.                      |
|  File Transfer (FTP, SFTP) | Files must be transferred without any loss or corruption.              |
|  Online Banking            | Transactions and sensitive data require high reliability and security. |
|  Database Connections      | Queries and responses must be accurate and complete.                   |
|  E-commerce & Payments     | Financial transactions require reliability and order.                  |
|  API Calls & Microservices | Requests and responses must be complete and in order.                  |

VS



#### REMEMBER

TCP = Accuracy  
& Reliability

UDP = Speed  
& Efficiency

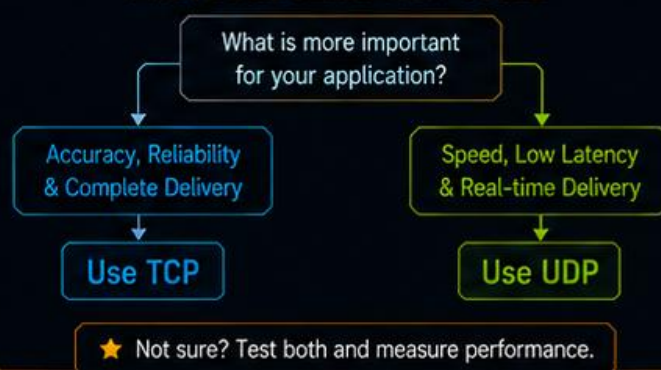
### WHEN TO USE UDP – REAL WORLD EXAMPLES

| APPLICATION / SCENARIO                                                                                               | WHY UDP IS THE RIGHT CHOICE                                 |
|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
|  Online Gaming                    | Low latency is critical. Small packet loss is tolerable.    |
|  Video Streaming (Live)           | Real-time delivery is more important than perfect delivery. |
|  VoIP & Video Calls               | Timely voice/video is critical. Minor loss is acceptable.   |
|  Live Broadcast / Streaming       | Speed to many users is essential, not perfect delivery.     |
|  DNS Lookups                      | Small queries/responses. Speed is more important.           |
|  IoT / Sensor Data                | Devices send small, frequent data with low latency.         |
|  Real-time Monitoring & Telemetry | Fast updates are critical, some data loss is acceptable.    |

### TCP vs UDP AT A GLANCE

| ASPECT         | TCP                                     | UDP                           |
|----------------|-----------------------------------------|-------------------------------|
| Connection     | Connection-oriented                     | Connectionless                |
| Reliability    | Reliable (ACKs, Retransmissions)        | Unreliable (No ACKs)          |
| Ordering       | Ensures Order                           | No Order Guarantee            |
| Speed          | Slower (More Overhead)                  | Faster (Less Overhead)        |
| Error Handling | Retransmits Lost Packets                | No Retransmission             |
| Header Size    | 20–60 Bytes                             | 8 Bytes                       |
| Best For       | Accuracy, Reliability, Ordered Delivery | Speed, Low Latency, Real-time |

### PROTOCOL SELECTION GUIDE



### BEST PRACTICES



#### Use TCP when:

- ✓ Data integrity is critical
- ✓ All data must be delivered
- ✓ Order of packets matters
- ✓ Security is a priority
- ✓ Examples: Web, Email, Banking, File Transfer, APIs



#### Use UDP when:

- ✓ Low latency is critical
- ✓ Some packet loss is acceptable
- ✓ Real-time delivery is more important than perfect delivery
- ✓ High speed and scalability are needed
- ✓ Examples: Gaming, Streaming, VoIP, DNS, IoT



### KEY TAKEAWAY

TCP gets it right. UDP gets it there fast. Choose based on what your application values most – perfect delivery or perfect timing.



Both have their strengths. Neither is better overall.



Right choice = better performance, lower cost and happier users.



Understand your application needs first, then choose your protocol.



The Internet works best when you use the right tool for the right job.



TCP and UDP power the Internet. Your choice makes the difference.

@manishburdak45



**THERE IS NO ONE-SIZE-FITS-ALL.**

The right protocol depends on what matters most for your application.



## TCP (Transmission Control Protocol)

Reliable, connection-oriented and feature-rich.  
Ensures data is delivered correctly and in order.



Connection  
Oriented



Reliable  
Delivery



Ordered  
Delivery



Error  
Recovery



Flow & Congestion  
Control



Higher  
Overhead

# VS



Connectionless



Best-Effort  
Delivery



Unordered  
Delivery



No Error  
Recovery



Low Overhead  
High Speed



Lightweight  
Header (8 Bytes)

## UDP (User Datagram Protocol)

Fast, connectionless and lightweight.  
Delivers data quickly with minimal overhead.

## WHEN TO USE?

### USE TCP WHEN...

- ✓ Accuracy, reliability and order are critical
- ✓ All data must be delivered completely
- ✓ Connection setup and retransmissions are acceptable
- ✓ Examples: Web, Email, File Transfer, Online Banking, APIs, etc.

### USE UDP WHEN...

- ✓ Speed and low latency are critical
- ✓ Some packet loss is acceptable
- ✓ Real-time data delivery is more important
- ✓ Examples: Streaming, Gaming, VoIP, Live Broadcast, IoT, DNS, etc.

## KEY DIFFERENCES

| ASPECT         | TCP                                     | UDP                           |
|----------------|-----------------------------------------|-------------------------------|
| Connection     | Connection-Oriented                     | Connectionless                |
| Reliability    | Reliable (ACKs, Retransmissions)        | Unreliable (No ACKs)          |
| Ordering       | Ordered Delivery                        | Unordered Delivery            |
| Speed          | Slower (More Overhead)                  | Faster (Less Overhead)        |
| Error Handling | Automatic Recovery                      | No Recovery                   |
| Header Size    | 20–60 Bytes                             | 8 Bytes                       |
| Best For       | Accuracy, Reliability, Ordered Delivery | Speed, Low Latency, Real-time |

## REAL-WORLD EXAMPLES

### TCP EXAMPLES



Web Browsing (HTTP/HTTPS)



Email (SMTP, IMAP, POP3)



Online Banking



File Transfer (FTP, SFTP)



API Calls & Microservices

### UDP EXAMPLES



Online Gaming



Video Streaming (Live)



VoIP & Video Calls



Live Broadcast / Streaming



IoT / Sensor Data



DNS Lookups

## PERFORMANCE IMPACT



- ✓ TCP ensures accuracy and reliability at the cost of speed and overhead.
- ✓ UDP delivers speed and low latency with minimal overhead but without guarantees.
- ✓ Test and measure to choose what matters most.

## BEST PRACTICES



- ✓ Understand your application requirements first.
- ✓ Use TCP for critical data and transactions.
- ✓ Use UDP for real-time and performance-critical applications.
- ✓ Test, monitor and optimize continuously.

## CHOICE GUIDE



Ask what is more important:

- ✓ Accuracy, Reliability & Complete Delivery?  
→ Use **TCP**
- ✓ Speed, Low Latency & Real-time Delivery?  
→ Use **UDP**

Not sure? Test both and decide.

## BOTTOM LINE



TCP and UDP both power the Internet. They are not competitors, they complement each other.

**The right choice depends on your application, your users and your goals.**



**CHOOSE WISELY. BUILD BETTER. DELIVER EXCELLENCE.**



Both **TCP** and **UDP** are powerful. The right choice makes all the difference.

@manishburdak45